

Assignment - Nearly Free Electron Model.

PAGE NO.:

TO SHOW 1- Bands will have zero slope at the middle & edges of the B-Zs.

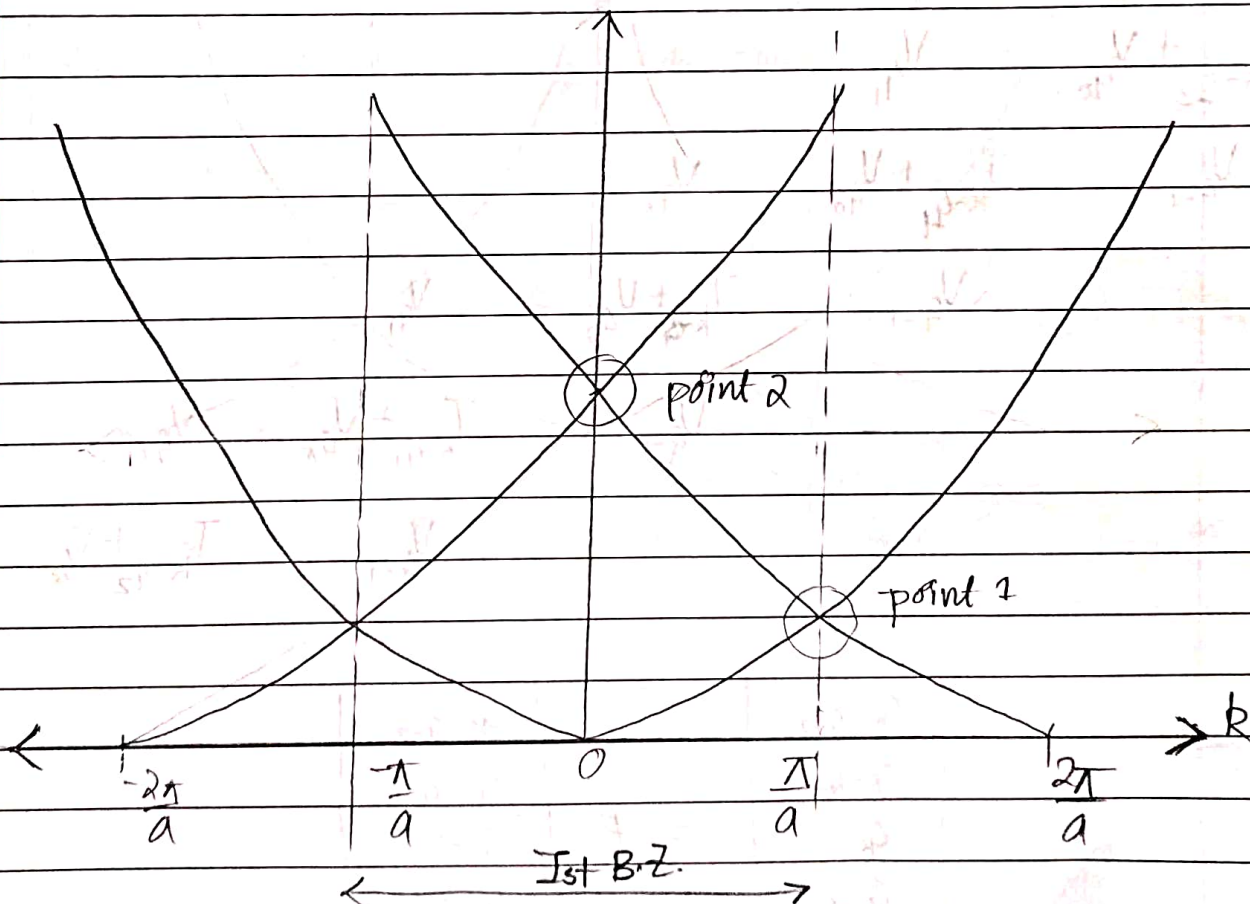
Consider a potential

$$V(x) = \sum_G V_G e^{iGx}$$

$$G = \frac{l2\pi}{a}$$

$$l \in \mathbb{Z}$$

We know the free electron band structure.
(before switching on the potential)
* keeping the periodicity of potential.



We will show that the bands have zero slope after switching on the potential feebly enough to be treated as perturbation at points 1 & 2.

At point 1:-

(mixing near the edge of B.Z.)

e^{ikx} , $e^{i(k-q_1)x}$ would mix at $k = \frac{\pi}{2}$.

Picking out the subspace spanned by

e^{ikx} & $e^{i(k-q_1)x}$.

$T_{k-q_2} + V_{q_0}$	V_{q_1}			
$V_{q_{-1}}$	$T_{k-q_1} + V_{q_0}$	V_{q_1}		
	$V_{q_{-1}}$	$T_{k-q_0} + V_{q_0}$	V_{q_1}	
		$V_{q_{-1}}$	$T_{k-q_1} + V_{q_0}$	V_{q_1}
			$V_{q_{-1}}$	$T_{k-q_2} + V_{q_0}$

c_{k-q_2}		c_{k-q_2}
c_{k-q_1}		c_{k-q_1}
c_k	$= E$	c_k
c_{k-q_1}		c_{k-q_1}
c_{k-q_2}		c_{k-q_2}



Apply 2nd order degenerate perturbation theory, we have

$$\begin{bmatrix} T_k + V_{q_0} & V_{q_1} \\ V_{q_{-1}} & T_{k-q_1} + V_{q_0} \end{bmatrix} \begin{bmatrix} C_k \\ C_{k-q_1} \end{bmatrix} = E \begin{bmatrix} C_k \\ C_{k-q_1} \end{bmatrix}$$

Diagonalizing

$$\begin{vmatrix} T_k + V_{q_0} - E & V_{q_1} \\ V_{q_{-1}} & T_{k-q_1} + V_{q_0} - E \end{vmatrix} = 0$$

$$\Rightarrow (T_k + V_{q_0} - E)(T_{k-q_1} + V_{q_0} - E) - V_{q_1} V_{q_{-1}} = 0$$

Taking V_{q_0} as reference potential and setting it to zero. (its just a constant which translates our band structure up & down)

$$\Rightarrow \cancel{T_k} (T_k - E)(T_{k-q_1} - E) - |V_{q_1}|^2 = 0$$

$$\Rightarrow T_k T_{k-q_1} + E^2 - E(T_k + T_{k-q_1}) - |V_{q_1}|^2 = 0$$

Quadratic formula,

$$\cancel{E} = \frac{1}{2} (T_k + T_{k-q_1}) \pm \sqrt{\left(\frac{T_k + T_{k-q_1}}{2}\right)^2 - \cancel{E}}$$

$$E = \frac{1}{2} (T_k + T_{k-q_1}) \pm \sqrt{\left(\frac{T_k + T_{k-q_1}}{2}\right)^2 - (T_k T_{k-q_1} - |V_{q_1}|^2)}$$

$$= \frac{1}{2} (T_k + T_{k-q_1}) \pm \sqrt{\left(\frac{T_k - T_{k-q_1}}{2}\right)^2 + |V_{q_1}|^2}$$

$$T_k = T_{k-q_1} \quad (\text{periodicity in } k\text{-space})$$

$$(@ k = \pi/a)$$

Hence we have

$$E = T_k \pm |V_{g_1}|$$

Hence, the 1st band is governed by the 1st harmonic of the potential.

We had,

$$E^2 = E(T_k + T_{k-g_1}) + |V_{g_1}|^2$$

Taking derivative wrt k ,

$$\Rightarrow 2E \frac{\partial E}{\partial k} - \frac{\partial E}{\partial k} (T_k + T_{k-g_1}) - E \left(\frac{\partial T_k}{\partial k} + \frac{\partial T_{k-g_1}}{\partial k} \right) = 0$$

$$\Rightarrow \frac{\partial E}{\partial k} [2E - T_k - T_{k-g_1}] - E \left(\frac{2\hbar^2 k}{2m} + \frac{2\hbar^2 (k-g_1)}{2m} \right) = 0$$

$$\Rightarrow \text{At } k = \frac{\pi}{a} \quad (\text{Also } T_k = T_{k-g_1})$$

$$\frac{\partial E}{\partial k} [2(T_k + |V_{g_1}|) - 2T_k] - E \frac{\hbar^2}{m} (2k - g_1) = 0$$

$$@ \quad k = \frac{\pi}{a} \quad g_1 = \frac{2\pi}{a} \quad \Rightarrow \quad 2k - g_1 = 0$$

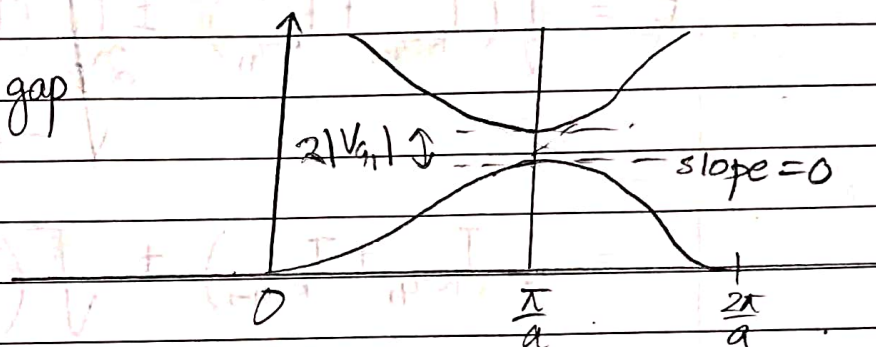
$$\Rightarrow \frac{\partial E}{\partial k} \cdot (2|V_{g_1}|) = 0$$

we know $|V_{q_1}| \neq 0$,

$$\text{Hence } \left. \frac{\partial E}{\partial k} \right|_{k=\frac{\pi}{a}} = 0 //$$

We proved that the slope goes to zero near the edge of the B.Z.

& the first band gap is governed by 1st harmonic.



At point -2 : (Middle of the B.Z.)

$e^{i(k-q_1)x}$ & $e^{i(k-q_1)x}$ would mix.

Consider the subspace spanned by $e^{i(k-q_1)x}$ & $e^{i(k-q_1)x}$ & applying degenerate perturbation theory @ $k=0$.

We have 2×2 block

$$\begin{bmatrix} T_{k-q_1} + V_{q_0} & V_{q_2} \\ V_{q_2} & T_{k-q_1} + V_{q_0} \end{bmatrix} \begin{bmatrix} c_{k-q_1} \\ c_{k-q_1} \end{bmatrix} = E \begin{bmatrix} c_{k-q_1} \\ c_{k-q_1} \end{bmatrix}$$

Taking $V_{q_0} = 0$ & Diagonalizing

$$\begin{vmatrix} T_{k-q_1} - E & V_{q_2} \\ V_{q_2} & T_{k-q_1} - E \end{vmatrix} = 0$$



$$\Rightarrow (T_{k-q_1} - E)(T_{k-q_1} - E) - |V_{q_2}|^2 = 0$$

$$\Rightarrow E^2 + T_{k-q_1} T_{k-q_1} - E(T_{k-q_1} + T_{k-q_1}) - |V_{q_2}|^2 = 0$$

Quadratic formula,

$$E = \frac{1}{2} (T_{k-q_1} + T_{k-q_1}) \pm \frac{1}{2} \sqrt{(T_{k-q_1} + T_{k-q_1})^2 - 4(T_{k-q_1} T_{k-q_1} - |V_{q_2}|^2)}$$

$$= \frac{1}{2} (T_{k-q_1} + T_{k-q_1}) \pm \sqrt{\left(\frac{T_{k-q_1} - T_{k-q_1}}{2}\right)^2 + |V_{q_2}|^2}$$

putting $T_{k-q_1} = T_{k-q_1}$ (periodicity in k -space)
(@ $k=0$)

$$E = T_k \pm |V_{q_2}|$$

$\therefore$ 2nd energy gap is governed by the second harmonic

we had

$$E^2 + T_{k-q_1} T_{k-q_1} - E(T_{k-q_1} + T_{k-q_1}) - |V_{q_2}|^2 = 0$$

Differentiate w.r't k ,

$$\frac{\partial E}{\partial k} \frac{\partial E}{\partial k} - \frac{\partial E}{\partial k} (T_{k-q_1} + T_{k-q_1}) - E \left(\frac{\partial T_{k-q_1}}{\partial k} - \frac{\partial T_{k-q_1}}{\partial k} \right) = 0$$

$$\Rightarrow \frac{\partial E}{\partial k} [2E - T_{k-q_1} - T_{k-q_2}] - E \left(\frac{2\hbar^2(k-q_1)}{2m} + \frac{2\hbar^2(k-q_2)}{2m} \right) = 0$$

At $k=0$,

$$T_{k-q_1} = T_{k-q_2}$$

$$q_1 = \frac{2\pi}{a} \quad q_2 = -\frac{2\pi}{a}$$

~~$k=q_1$~~

$$\Rightarrow \frac{\partial E}{\partial k} [2(T_k + |V_{q_2}|) - 2T_k] - \frac{E\hbar^2}{m} (2k - q_1 - q_2) = 0$$

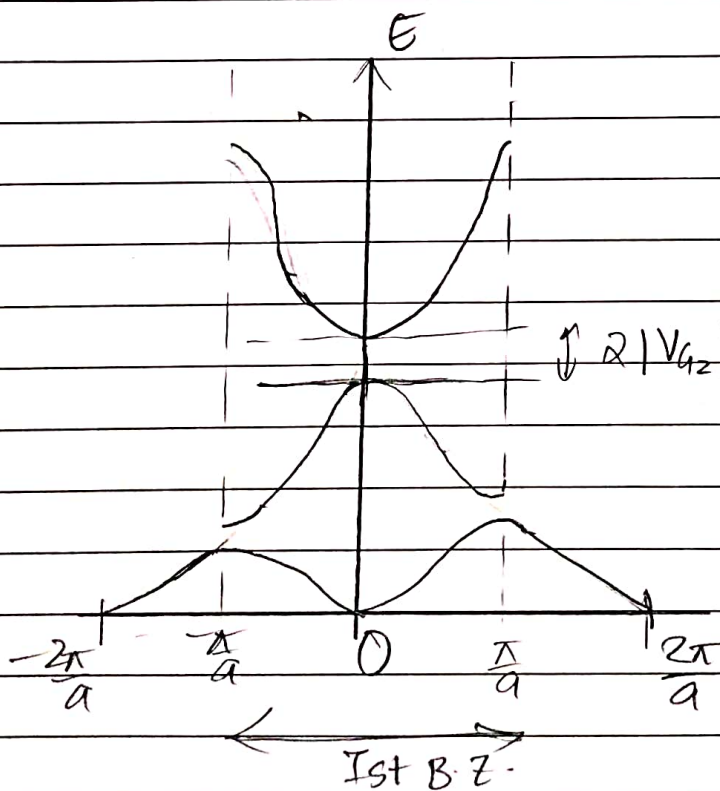
$$2k - q_1 - q_2 = 0 + \frac{2\pi}{a} - \frac{2\pi}{a} = 0$$

$$\Rightarrow \frac{\partial E}{\partial k} [2|V_{q_2}|] = 0$$

we know $|V_{q_2}| \neq 0$

Hence

$$\left[\frac{\partial E}{\partial k} \right]_{k=0} = 0$$



$\uparrow 2|V_{q_2}|$ & slope = 0 //

Hence, we conclude that band gaps depend on the strength of corresponding harmonics &

have zero slope near the edge & middle of Brillouin zones.